A worked example for $PGL(2)$

Draft section for the 2026 revision of D. Joyner,
"Simple local trace formulas for unramified p-adic groups" (1997)

Notes prepared by Claude (Anthropic) for D. Joyner

November 10, 2026

Abstract

For $G = PGL(2)$ over a p-adic field we compute, in closed form, the distribution $I(\chi, \gamma)$ of the main identity

$$\int_{T_\gamma \backslash G(F)} f(g^{-1}\gamma g)\, dg = \int f^\vee(\chi)\, I(\chi, \gamma)\, d\mu(\chi) \qquad (f \in \mathcal{H}(G, K),\ \gamma \in G_{ell})$$

for every type of regular elliptic γ. The computation uses the Bruhat–Tits tree on the geometric side and Macdonald's formula on the spectral side. Every formula was checked numerically for $q = 3$, against a direct count of vertices on the tree and by numerical integration over the unitary circle. The example confirms the corrected normalizations in the referee report (M1). It shows that $I(\chi, \gamma)$ is a rational function of χ whose poles at the non-regular characters are cancelled by the Plancherel density. It also shows that the dependence on the truncation parameter is a "vertex term" that pairs to zero *exactly* with $f^\vee$ once the truncation is large. That last point is the model for the Route A repair of §4.

1 Setup and conventions

Let F be a p-adic field with ring of integers $\mathcal{O}$, uniformizer π and residue field of order q. Let $G = PGL(2)$, $K = PGL(2, \mathcal{O})$, and let A be the diagonal torus (modulo the center). Then $X_*(A) = \mathbb{Z}$, generated by the fundamental coweight $\varpi^\vee$ with $\varpi^\vee(\pi) = a_1$, where

$$a_n := \mathrm{diag}(\pi^n, 1) \qquad (n \in \mathbb{Z}).$$

The simple coroot is $\alpha^\vee = 2\varpi^\vee$, so $a_\alpha = \alpha^\vee(\pi) = a_2$ modulo the center. The group G is adjoint, which is why $X_*(A) \neq \mathbb{Z}\alpha^\vee$ here.

Characters, Weyl group, chamber. An unramified character χ of $A(F)$ is determined by $z = \chi(a_1) \in \mathbb{C}^\times$; it is unitary iff $|z| = 1$. Then $\chi(a_n) = z^n$ and $\chi(a_\alpha) = z^2 =: y$. The Weyl group $W = \{1, s\}$ acts by $z \mapsto z^{-1}$. The chamber $A^- = \{a : |\alpha(a)| \leq 1\}$ of the paper is $\{a_n : n \geq 0\}$, since $|\alpha(a_n)| = |\pi^n|$. We have $\delta_B(a_n)^{1/2} = q^{-n/2}$.

Measures. We take $\mathrm{vol}(K) = 1$. The centralizer T_γ of a regular elliptic γ is compact, and we give it volume 1. Then

$$\mathcal{O}_\gamma(f) := \int_{T_\gamma \backslash G} f(g^{-1}\gamma g)\, dg = \int_G f(g^{-1}\gamma g)\, dg.$$

The double cosets are Ka_nK, $n \geq 0$, with $\mathrm{vol}(Ka_nK) = (q+1)q^{n-1}$ for $n \geq 1$. Let $f_n := \mathbf{1}_{Ka_nK}$.

2 The spherical side

Macdonald's formula. Let $c_\alpha(\chi) = \dfrac{1 - q^{-1}\chi(a_\alpha)}{1 - \chi(a_\alpha)}$ be as in (2.8) of the paper (split case). For $n \geq 0$,

$$\Gamma_\chi(a_n) = Q^{-1} q^{-n/2} \Big(c_\alpha(\chi^{-1})\, z^n + c_\alpha(\chi)\, z^{-n} \Big), \qquad Q = 1 + q^{-1}. \tag{1}$$

Two features of this formula differ from Lemma 2.1 as printed:

- $Q = \sum_w q^{-\ell(w)}$, not $\sum_w (IwI : I)$ (report M1(a)).
- On the paper's chamber A^-, the coefficient of $(w\chi)(a)$ is $c(w\chi^{-1})$, not $c(w\chi)$. In other words, Lemma 2.1 as printed is Macdonald's formula on the *opposite* chamber.

Checks: at $n = 0$ the bracket equals $1+q^{-1}$, so $\Gamma_\chi(1) = 1$. Formula (1) was also verified numerically to machine precision.

Satake transform and Plancherel measure. The Satake transform of f_n is

$$f_n^\vee(\chi) = \mathrm{vol}(K a_n K)\, \Gamma_\chi(a_n).$$

On the unitary circle $|z| = 1$ (with $d\theta/2\pi$, $z = e^{i\theta}$), the Plancherel density is

$$d\mu(\chi) = \rho(z)\, \frac{d\theta}{2\pi}, \qquad \rho(z) = \frac{Q}{|W|}\, |c_\alpha(\chi)|^{-2} = \frac{Q}{2}\, \frac{|1 - y|^2}{|1 - q^{-1} y|^2}, \qquad y = z^2. \tag{2}$$

Proposition 1 (Plancherel inversion; Corollary 2.3)**.** *For all $n, m \geq 0$,*

$$\int_{|z|=1} f_n^\vee(\chi)\, \Gamma_\chi(a_m)\, d\mu(\chi) = \delta_{nm} = f_n(a_m).$$

This was checked numerically for $0 \leq n, m \leq 3$ (Table 1). With the printed constant $Q = 1 + q$, the diagonal entries would be q instead of 1.

3 The geometric side: counting on the tree

Let $\mathcal{X}$ be the Bruhat–Tits tree of G (vertices G/K, regular of degree $q+1$) and let d be its graph metric. Since f_n is bi-K-invariant, $g \mapsto f_n(g^{-1}\gamma g)$ is right K-invariant, and $g^{-1}\gamma g \in K a_n K$ iff $d(gK, \gamma g K) = n$. Hence

$$\mathcal{O}_\gamma(f_n) = \#\{ v \in \mathcal{X} :\ d(v, \gamma v) = n \}. \tag{3}$$

The set is finite because γ is elliptic.

Up to conjugacy and the center, a regular elliptic γ generates $E = F[\gamma]$, a quadratic extension. There are three types:

($\mathbf{U}_r$) E/F unramified, $\gamma \in \mathcal{O}_E^\times$, of depth $r \geq 0$. The fixed-point set is the ball $B_r(v_0)$ of radius r about a vertex. Each vertex at distance $m \geq 1$ from the ball has $d(v, \gamma v) = 2m$.

($\mathbf{R}_r$) E/F ramified, $\gamma \in \mathcal{O}_E^\times$, of depth $r \geq 0$. The fixed-point set is the set of vertices within distance r of an edge e. Each vertex at distance $m \geq 1$ from it has $d(v, \gamma v) = 2m$.

($\mathbf{R}'$) E/F ramified and $v_E(\gamma)$ odd (e.g. $\gamma = \pi_E$). Then γ reverses an edge e, and $d(v, \gamma v) = 2\, d(v, e) + 1$.

Counting vertices in spheres gives:

Proposition 2. *Let $m \geq 1$.*

type	$\mathcal{O}_\gamma(f_0)$	$\mathcal{O}_\gamma(f_{2m})$	$\mathcal{O}_\gamma(f_{2m+1})$
(U_r)	$\lvert B_r\rvert = 1 + \dfrac{(q+1)(q^r-1)}{q-1}$	$(q+1)q^{r+m-1}$	0
(R_r)	$\dfrac{2(q^{r+1}-1)}{q-1}$	$2q^{r+m}$	0
(R')	0	0	$2q^m$ *(also $\mathcal{O}_\gamma(f_1)=2$)*

These counts were verified for $q = 3$ by enumerating all 4373 vertices of the ball of radius 7 in the tree of $PGL(2, \mathbb{Q}_3)$, with γ equal to

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathrm{U}_0), \quad \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix} (\mathrm{U}_1), \quad \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} (\mathrm{R}_0), \quad \begin{pmatrix} 0 & 3 \\ 1 & 0 \end{pmatrix} (\mathrm{R}').$$

See Table 2.

4 The spectral side: $I(\chi, \gamma)$ in closed form

Write $f = \sum_n f(a_n) f_n$ (a finite sum), so $\mathcal{O}_\gamma(f) = \sum_n f(a_n)\mathcal{O}_\gamma(f_n)$. Insert Proposition 1 for $f(a_n)$. For any $N \geq \deg f$ (the largest n with $f(a_n) \neq 0$) this gives the *exact* identity

$$\mathcal{O}_\gamma(f) = \int f^\vee(\chi)\, I^N(\chi, \gamma)\, d\mu(\chi), \qquad I^N(\chi, \gamma) := \sum_{n=0}^{N} \mathcal{O}_\gamma(f_n)\, \Gamma_\chi(a_n). \tag{4}$$

This is the paper's truncated distribution I^T of §4.2, with $T \leftrightarrow N$. Now let $N \to \infty$. Write $y = z^2$ and $\tilde{\theta}(\chi) = 1 - \chi(a_\alpha) = 1 - y$.

Theorem 3. *For regular unitary χ (i.e. $z \neq \pm 1$), the limit $I(\chi, \gamma) = \lim_N I^N(\chi, \gamma)$ exists as an Abel limit. It is the rational function*

$$\textit{(U_r):} \quad I(\chi,\gamma) = \lvert B_r\rvert \;+\; q^r \sum_{w \in W} c_\alpha\big((w\chi)^{-1}\big)\, \frac{(w\chi)(a_\alpha)}{\tilde{\theta}(w\chi)} \;=\; \lvert B_r\rvert \;+\; q^r\, \frac{\cos\theta - q^{-1}}{1 - \cos\theta} \qquad (y = e^{i\theta}),$$

$$\textit{(R_r):} \quad I(\chi,\gamma) = \frac{2(q^{r+1}-1)}{q-1} \;+\; \frac{2q^{r+1}}{q+1} \cdot \frac{\cos\theta - q^{-1}}{1 - \cos\theta},$$

$$\textit{(R'):} \quad I(\chi,\gamma) = \frac{q^{1/2}(1-q^{-1})}{q+1} \cdot \frac{\cos\psi}{\sin^2\psi} \qquad (z = e^{i\psi}).$$

Moreover, $\mathcal{O}_\gamma(f) = \int f^\vee(\chi) I(\chi, \gamma)\, d\mu(\chi)$ for every $f \in \mathcal{H}(G, K)$, and the integral converges absolutely.

Sketch. Consider type (U_r); the other types are identical. By Proposition 2 and (1), the coefficient $\mathcal{O}_\gamma(f_{2m})\, Q^{-1} q^{-m}$ equals q^r for every $m \geq 1$. So

$$I^N = \lvert B_r\rvert + q^r \sum_{m=1}^{N/2} \big(c_\alpha(\chi^{-1}) y^m + c_\alpha(\chi) y^{-m}\big).$$

Summing the geometric series gives the closed form, plus the remainder

$$I^N - I = -q^r\Big(c_\alpha(\chi^{-1})\frac{y^{M+1}}{1-y} + c_\alpha(\chi)\frac{y^{-M-1}}{1-y^{-1}}\Big), \qquad M = \lfloor N/2 \rfloor.$$

Multiply by the Plancherel density, using $\rho = \frac{Q}{2}\,(c_\alpha(\chi)c_\alpha(\chi^{-1}))^{-1}$ on $|z| = 1$:

$$(I^N - I)\,\rho = -\frac{Q}{2}\,q^r\Big(\frac{y^{M+1}}{1-q^{-1}y} + \frac{y^{-M-1}}{1-q^{-1}y^{-1}}\Big).$$

The first term has only Fourier modes y^k with $k \geq M+1$, the second only $k \leq -M-1$. Since $f^\vee$ is a Laurent polynomial in $y^{\pm 1/2} = z^{\pm 1}$ of degree $\leq \deg f$, we get $\int f^\vee (I^N - I)\,d\mu = 0$ *exactly* once $2(M+1) > \deg f$. Combine this with (4). Absolute convergence holds because $I\rho$ is smooth; see Remark 4. □

Remark 4 (Poles and the Plancherel density). $I(\chi, \gamma)$ has a *double* pole at the non-regular characters $z = \pm 1$ ($w\chi = \chi$), and no other poles on the unitary circle. The Plancherel density ρ has a *double zero* there. Their product is smooth, and in fact

$$I(\chi,\gamma)\,\rho(z) = \frac{Q}{2}\cdot\frac{|B_r|\,(2-2\cos\theta) + q^r\,(2\cos\theta - 2q^{-1})}{1 - 2q^{-1}\cos\theta + q^{-2}} \qquad (\text{type } \mathrm{U}_r).$$

So in rank one, "rational function" (1997 introduction) is right, *provided* it is always paired with $d\mu$. On its own, $I(\cdot, \gamma)$ is not locally integrable at $z = \pm 1$ against $d\theta$, which may explain why the 1992 abstract said "distribution."

The cancellation is structural. For each w,

$$\frac{c_\alpha((w\chi)^{-1})\,\rho}{\tilde\theta(w\chi)} = \frac{Q}{2}\cdot\frac{1}{1 - q^{-1}(w\chi)(a_\alpha)},$$

because $\rho \propto |c|^{-2}$ supplies the factor $1-(w\chi)(a_\alpha)$ that cancels $\tilde\theta$. In higher rank the same mechanism cancels the factors of $\tilde\theta$ that vanish on walls. Whether cone denominators that vanish *off* the walls also cancel, after summing over w, w', is exactly the $(G, A)^\sim$-family statement of §2.5. That is the one genuinely new ingredient Route A needs in rank ≥ 2.

Remark 5 (What the example says about the truncation parameter). Equation (4) shows that $f^\vee \mapsto \int f^\vee I^N\,d\mu$ is independent of N once $N \geq \deg f$. That is automatic, and it is the first line of the paper's Theorem 4.5. What actually needs proof is *regularity*: that $\lim_N I^N$ exists as a function and that $I\rho$ is integrable. The dependence on N is a single "vertex term" $y^{M+1}\cdot$(holomorphic in $|y| < q$), together with its W-conjugate, so it pairs to zero exactly, with no Riemann–Lebesgue argument needed. This suggests restating Proposition 4.2 and Theorem 4.5 as follows. Define $I(\chi, \phi)$ as the T-independent (vertex 0) part of the Brion decomposition of I^T. Then (i) $I^T - I$ is a sum of vertex terms $\chi(T_\omega)R_\omega(\chi)$, and (ii) each $R_\omega\rho$ is smooth with Fourier support in a translated cone. Together these give $\int f^\vee(I^T - I)\,d\mu = 0$ for $T \geq T(f, \phi)$.

Remark 6 (The "core" term). In type (U_r) the constant $|B_r| = \mathcal{O}_\gamma(\mathbf{1}_K)$ comes from the vertices fixed by γ. Those are exactly the points where γ acts without displacement, so no Casselman asymptotic is available there. It is non-zero and depends on the depth r, while the second (cone) term has the shape predicted by Proposition 4.2: a sum over W of c-functions over $\tilde\theta$. This is the concrete form of the gap noted in the log (item 3.69): any general argument must account for such a term, and the induction in Proposition 3.6 does not produce it. For type (R') the core is empty, and I is a pure cone term.

Remark 7 (Shape and sign). $I(\chi,\gamma)$ depends on γ only through its type and depth, as it must. In types (U_r) and (R_r), I is positive on the regular unitary characters. Its minimum is at $\cos\theta=-1$ ($z=\pm i$), where for (U_r) it equals $|B_r|-q^r\frac{1+q^{-1}}{2}>0$, and $I\to+\infty$ as $z\to\pm1$. In type (R'), I changes sign with $\cos\psi$. So $I(\cdot,\gamma)\,d\mu$ is a signed measure in general, not a positive one.

5 Numerical verification ($q=3$)

All numbers below were produced by the script `pgl2_check.py`, which accompanies this note. Orbital integrals are exact counts on the tree; spectral integrals use the midpoint rule with 4×10^5 nodes on $|z|=1$.

$\int f_n^\vee\,\Gamma_\chi(a_m)\,d\mu$	$m=0$	1	2	3
$n=0$	1.0000	0.0000	0.0000	0.0000
$n=1$	0.0000	1.0000	0.0000	0.0000
$n=2$	0.0000	0.0000	1.0000	0.0000
$n=3$	0.0000	0.0000	0.0000	1.0000

Table 1: Plancherel inversion with $Q=1+q^{-1}$ (Proposition 1).

	(U_0)		(U_1)		(R_0)		(R')	
n	tree	spectral	tree	spectral	tree	spectral	tree	spectral
0	1	1.000000	5	5.000000	2	2.000000	0	0.000000
1	0	0.000000	0	0.000000	0	0.000000	2	2.000000
2	4	4.000000	12	12.000000	6	6.000000	0	0.000000
3	0	0.000000	0	0.000000	0	0.000000	6	6.000000
4	12	12.000000	36	36.000000	18	18.000000	0	0.000000
5	0	0.000000	0	0.000000	0	0.000000	18	18.000000
6	36	36.000000	108	108.000000	54	54.000000	0	0.000000
7	0	0.000000	0	0.000000	0	0.000000	54	54.000000

Table 2: $\mathcal{O}_\gamma(f_n)$ counted on the tree versus $\int f_n^\vee I(\chi,\gamma)\,d\mu$ with I from Theorem 3.

We also checked the following, all to machine precision:

- the trigonometric forms in Theorem 3;
- the formula for $I\rho$ in Remark 4;
- the chamber convention in (1).

For the truncation (Remark 5) with $f=f_4$ and type (U_0): $\int f_4^\vee I^N\,d\mu$ equals 0 for $N<4$ and exactly $12=\mathcal{O}_\gamma(f_4)$ for all $N\geq4$.

6 Consequences for the revision

1. **Normalizations.** Fix Lemma 2.1 (constant Q, chamber A^- with $c(w\chi^{-1})$) and Lemma 2.2 accordingly. Add these items to the log.

2. **Theorem 4.5.** State it with $I(\chi, \phi)$ defined as the T-independent part, and prove it by the exact-vanishing argument of Remark 5, not by "independence of T from Proposition 3.6."

3. **Regularity.** In rank one, $I\, d\mu$ is smooth. In general this needs: (a) the cancellation of wall factors (automatic, Remark 4); and (b) the $(G, A)^{\sim}$-family argument for cone denominators off the walls. Part (b) is the main step of Route A. A rank-two numerical test ($PGL(3)$, on the building) would be a good next check before writing it in general.

4. **Core term.** Proposition 3.6 must be restated as "explicit cone terms + a T-independent remainder." The example shows that the remainder is non-zero and geometric: $\mathcal{O}_\gamma(\mathbf{1}_K)$ in rank one.

5. **New section.** This example can go into the revision as a new section (or appendix), "The case of $PGL(2)$," with the tables.